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Application of a Rolling Bearing Life Model with Surface and Subsurface Survival to Hybrid Bearings in High-Load and High-Speed Applications

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Abstract

A previously developed and validated rolling bearing life model that separates the surface and the subsurface survivals is now adapted to account for superior bearing steel that allows bearings to operate in the region near contact pressures of 3 GPa and speeds of $nd_m = 3 \times 10^6$ (n is the rotational speed in rpm and d_m is the bearing mean diameter in mm), common of high-load and high-speeds applications like modern spindle bearings. The adapted model is verified with the support of endurance testing and its range of application is also verified with the support of a full non-seizure test. The adapted model can now be used in the bearing selection process for those tough application conditions.

Keywords

rolling bearing life, high-speed bearings, high-load bearings, seizure, fatigue, surface life

1 Introduction

Many modern applications of rolling bearings require very high speeds and very high loads, often combined with poor lubrication conditions or/and high solid contamination. Examples of these are modern spindles, electrical vehicle gearboxes and aero-engines, where rolling bearings need to survive very tough conditions. It has been known that hybrid rolling bearings (bearings made of steel rings and ceramic rolling elements of Si_3N_4) are more suitable for these kind of application conditions [1, 2]. However, until recently there were no bearing life models for hybrid rolling bearings where a comparison of performance of these bearings with steel-steel solutions could be made. Recently, within the same research group from the authors a life model for hybrid rolling bearing was published [3] including full validation with endurance testing and also a comparative investigation between steel-steel and hybrid bearings [4]. These studies show the enormous advantages of hybrid bearings in poor lubrication conditions and in the presence of solid contamination in comparison with steel-steel solutions. The separation of surface and subsurface survival [5] also explains why if the life is dominated by tribological aspects at the surface then the hybrid bearing will have an advantage respect to the steel-steel solution, despite of a hybrid bearing having larger subsurface stresses due to its higher stiffness in the ceramic rolling elements. The fact that a bearing life model can separate the surface and subsurface

survival makes the prediction of this effect possible, showing substantial advantages compared to subsurface bearing life models that will always penalise the use of a hybrid bearing [6, 7] due to the higher subsurface stresses at the same load.

Now, hybrid bearings used in high-load and high-speed applications like the ones considered here require special attention, since this combination might induce seizure failures in the contact surfaces potentially assisted by kinematic starvation [8, 9]. Therefore if a bearing life model is intended to be used in these conditions, then a transition diagram separating the surface/subsurface fatigue failure modes from the seizure failure mode is required to avoid using a fatigue-based life model in an area where seizure is expected. Alternatively, the seizure failure mode would need to be integrated in the life model [10]. In the current study a transition diagram will be pursued. For this, minimum-life endurance tests are carried out under high-speed and high-load conditions. Besides this, the methodology for life calculation will be applied as a showcase to a new tougher steel based on powder metallurgy (here it will be referred as PM1) used to manufacture the hybrid bearings [9, 11]. Hence also full fatigue endurance tests to measure the fatigue strength of the bearings are carried out to validate the life model with this material.

Therefore, the objective of the present paper is twofold:

To re-adjust the earlier developed fatigue-based bearing life model [3] for application to hybrid bearings made from a tougher powder metallurgy steel (PM1).

methods. Endurance test rigs used are the SKF R-2 rigs, described elsewhere [14] but a simple layout is shown in Fig. 2.

The rigs are arranged in batteries, each battery has 10 or 15 R-2 rigs capable of simultaneously testing up to 30 bearings. All the bearings in each battery are lubricated with a dedicated oil lubrication system in order to achieve homogeneous conditions in all tested bearings. The lubrication system has a 600 litres oil reservoir, filtering and temperature conditioning. By controlling the oil temperature and the oil flow (cooling) to the bearings different lubrication qualities (κ) can be achieved in the tested bearings. During the test the load is hydraulically applied and it is monitored with the use of load cells. Cleanliness of the lubricant is controlled by a multi-pass filtration system of high capacity down to 3 microns pore size. Oil contamination levels during testing can be monitored continuously with on-line automatic particle counters also supported by spectrographic and SEM-EDAX (Scanning Electron Microscope) analysis.

Post-test investigations can be done with the use of microscopic and metrological examination of all failures with categorization of the failure type, origin and bearing component that is affected. Usually, all failures, either originated from the surface, subsurface or from other critical component of the bearing, are included in the fatigue life analysis.

The hybrid bearings endurance tested reported here are angular contact ball bearings of designation 7209 with ISO internal geometry and a nominal contact angle of 25 deg. The bearing rings are made of PM1 powder metallurgy steel in bars with basic information and composition given in [9, 11] and the rolling elements of bearing grade silicon nitride (Si_3N_4). The bearings were all manufactured and heat treated in-house. The bearings were axially loaded with 30 kN load each, which gives a nominal maximum Hertzian contact pressure of 3.2 GPa in the inner ring contact and 2.76 in the outer ring contact. Figure 3 shows the contact pressures calculated using an internal software for the inner ring (IR) and the outer ring (OR) contacts. These show a small contact truncation of around 13% and 7%,

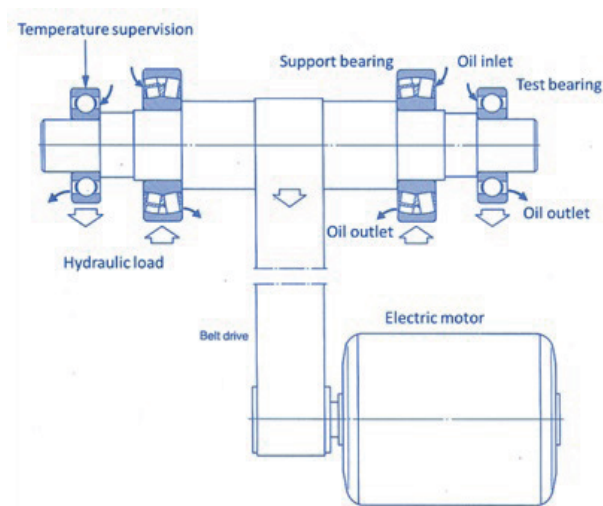


Fig. 2 Basic design of the SKF-R2 test rig

respectively, which should not influence the bearing life. A summary of the application conditions of the test is given in Table 1. The tests were carried out under clean oil conditions.

Two populations of 20 and 22 hybrid bearings each were tested, the first one with bearing rings made of standard ASTM 52100 steel and the second one of the PM1 steel to allow for a clear comparison in performance. The obtained bearing lives are normalised using as reference the shortest measured life in the baseline test with ASTM 52100 steel. A summary of the two endurance test results are shown as a Weibull probability plot in Fig. 4, where it can be seen that the increase of the $L_{10,50}$ for the steel PM1 is roughly a factor 15 compared to the $L_{10,50}$ of the baseline ASTM 52100 steel. Of course, the L_{10} life for the former also shows larger confidence intervals (poorer statistics) due to the low number of failures registered in the test (only 2). This needs to be taken into consideration when adapting the model to the new material by taking a more conservative calculation. In fact the testing time with the new material PM1 was very long and it was very difficult to make the bearings fail (15 times longer than expected), thus at some point the testing team decided to stop the test since with the two obtained failures (although they do not give very narrow confidence intervals) the separation respect to the reference test confidence intervals is very clear, thus it is possible to accept these results despite the slightly bigger confidence intervals. The reader should also notice that in [4] a much larger test data set is presented to validate the basic hybrid bearing life model.

Under the given conditions of high load and reduced lubrication, failures were expected to be present in both surface and subsurface. All failed bearings were inspected after the test and indeed this was the case. Failures were present in both inner rings and outer ring raceways. Neither ball nor cage failures were observed.

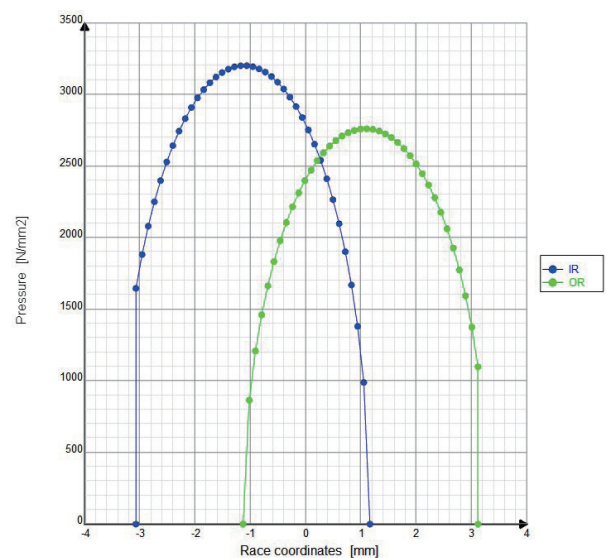


Fig. 3 Calculated contact pressures in the inner ring (IR) and outer ring (OR) contacts of the 7209 bearing according to the test operating conditions as summarised in Table 1

Table 1 Endurance test operating conditions

Bearing designation	Axial Force, [kN]	C/P	Oil viscosity at 40°C, [cSt]	Bearing temperature, [°C]	Rotational speed, [rpm]	Lubrication quality, κ , [-]	Approx. Δ value, [-]
7209	30	1.57	68	83	10000	2.6	2

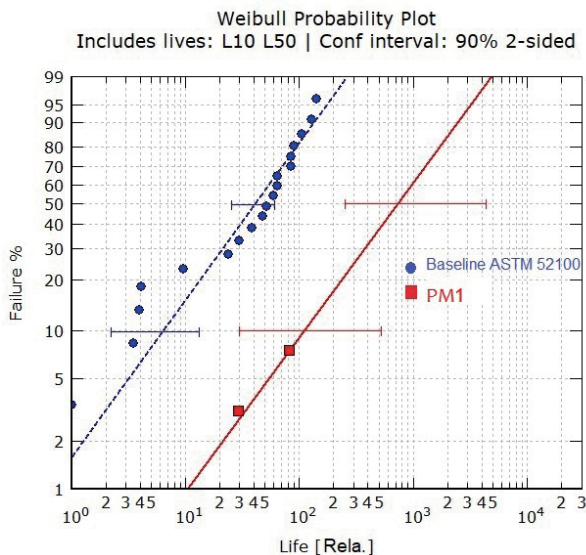


Fig. 4 Weibull plot for the hybrid bearings endurance tested under the conditions of Table 1. Measured lives are normalised using as reference the shortest measured life in the baseline test. The blue circles show bearing lives for the baseline test with steel ASTM 52100 and the red squares show bearing lives with the new PM1 steel. Confidence intervals 90% 2-sides are shown for L_{10} and L_{50} . Thus $L_{10,95}$ and $L_{10,5}$ and $L_{50,95}$ and $L_{50,5}$.

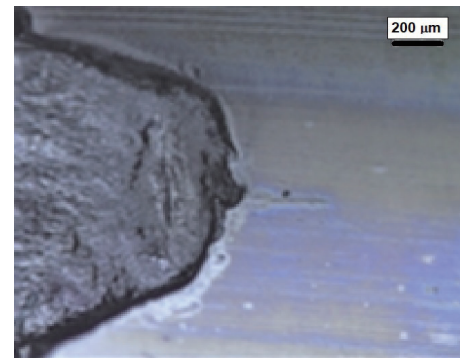
As described in [5] the current model can calculate the ratio of surface damage, which represents the probability of failures from the surface, this parameter can be calculated with the ratio of the surface damage integral and the subsurface one from Eq. (5). When doing this calculation for the test conditions 57% probability of surface failures is obtained, this number is very close to equal probability of failure for the surface and the subsurface, and indeed the test results reflect nearly the same proportion of surface and subsurface failures.

Figure 5 shows a set of pictures of representative failures in the bearings, selected from the two populations tested, showing surface and subsurface initiated spalls in both inner and outer rings.

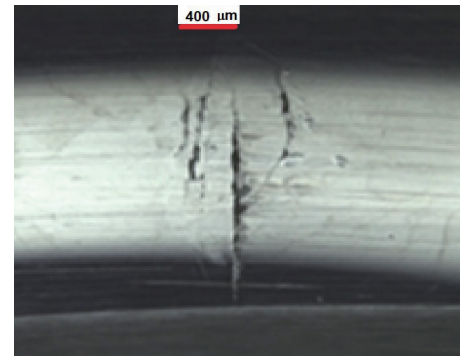
4 Non-seizure minimum-life test

The non-seizure minimum-life test has the objective to verify that under the selected operational conditions a bearing population can attain a certain minimum life with enough confidence. In other words, it verifies that the bearings do not show sudden and catastrophic failure modes, in this case seizure. The test is then performed for a small bearing population but it should have zero failures. In the present case, this test was carried out at the University of Aachen with an smaller hybrid bearing than the one used in the fatigue endurance test, and with a nominal contact angle of 20 deg. A full report of this test will be published in a separate publication, here only the relevant results are considered for the purposes already described. A summary of the application conditions for this test are given in Table 2. The calculated contact pressures are shown in Fig. 6.

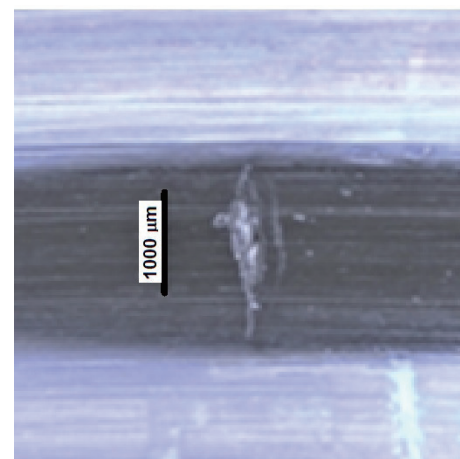
The design of the non-seizure test is based on the minimum-life concept which assumes a known Weibull slope value, so it reduces to the exponential probability distribution. In this case 1.1 was assumed (a value typically observed in rolling



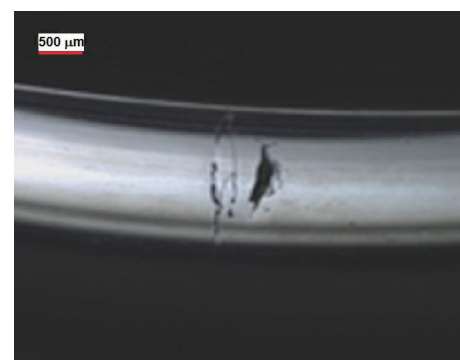
(a) IR Surface initiated (ASTM 52100)



(b) OR Subsurface initiated (ASTM 52100)



(c) IR Surface initiated (PM1)



(d) OR Subsurface initiated (PM1)

Fig. 5 Examples of typical failures found in analysed bearings from the two endurance tests

Table 2 Non-seizure minimum life test operating conditions

Bearing designation	Axial Force, [kN]	C/P	Oil viscosity at 40°C, [cSt]	Bearing temperature, [°C]	Rotational speed, [rpm]	Lubrication quality, κ , [-]	Approx. Λ value, [-]
7008	3	4.0	68	65	46000	4.4	3.1

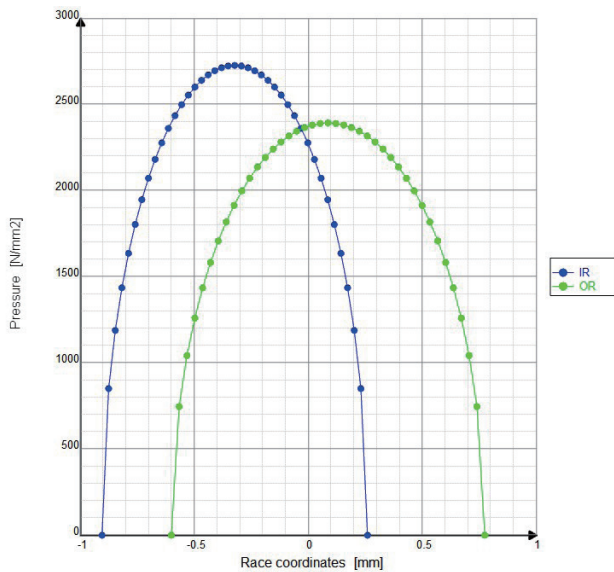


Fig. 6 Calculated contact pressures in the inner ring (IR) and outer ring (OR) contacts of the 7008 bearing according to the test operating conditions as summarised in Table 2

bearing test results). The calculated L_{10} value under the test conditions is known from the model, so the unknown is the number of bearings that need to survive this L_{10} value. If the test is successful it means that with 90% confidence the true life of the bearings is at least L_{10} . In the present case it turned out to be testing 6 bearings for 432 hours each with no failures. This test was successful and the operating conditions are validated for minimum life. The results can be summarised in Fig. 7

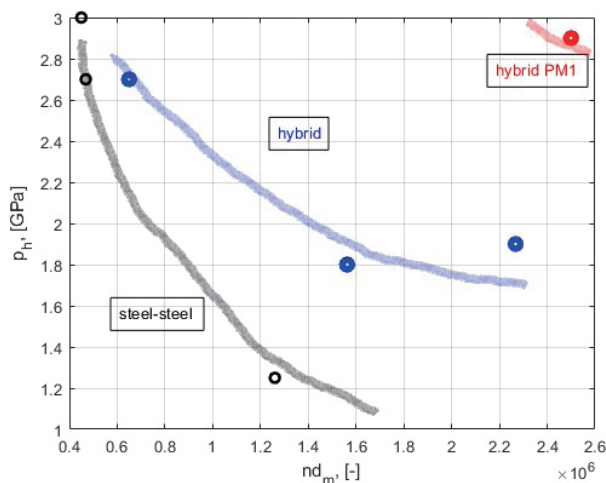


Fig. 7 Estimated transition diagram for the hybrid bearings made of PM1 steel. The conditions of the actual minimum life endurance test (non-seizure test) are marked in the diagram. For completeness the results obtained before for ASTM 52100 hybrid and steel-steel bearings from reference [8] have been added.

where the red circle represents the load-speed conditions from the current test. This is plotted including previous tests [8] for ASTM 52100 hybrid bearings and steel-steel bearings. The sketched lines added represent imaginary limits of a transition diagram, as explained in [7], beyond which the calculated fatigue life cannot be assured because the bearings could fail from seizure.

5 Comparisons with the model

The hybrid bearing life model briefly described above and presented in detail in [3] was modified for the new material PM1 according to the results from the endurance tests shown in Fig. 4. The L_{10} predictions of this model are shown in Fig. 8 with a red star symbol together with the test results. Since the test involved hybrid bearings, the surface model (second term of Eq. (5)) was not modified, the same model as [3] was used. The new material has two main performance parameters that needed to be covered by the model: (i) better fatigue strength (reflected in the subsurface) with a modification on the constant $\bar{A}$, the fatigue limit $\sigma_{u,v}$ was not modified and (ii) better resistance to seizure (reflected in higher application pressure-speed limit in Fig. 7). The other constants in Eq. (5) are the same used in engineering practice as mentioned in [5, 7] for radial ball bearings, i.e. $e = 10/9$, $c = 31/3$, $h = 2.33$.

For comparison, the prediction of the original model is also depicted with a blue star symbol and can be compared with the baseline test results.

It can be seen that the prediction of the model for the baseline test is very close to the $L_{10,50}$ value, which shows a moderate confidence. However, the model was adjusted to include higher safety for the PM1 material, with predictions

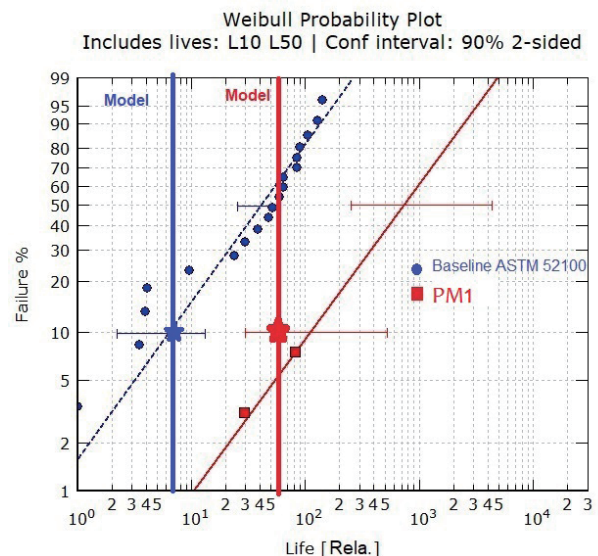


Fig. 8 Weibull probability plot for the endurance test lives results with superimposed model calculated (stars) L_{10} values for both, the baseline material ASTM 52100 and the new PM1 material

between $L_{10,95}$ and $L_{10,50}$, in fact the closest to the limit $L_{10,95}$ the more conservative the prediction will be.

6 Discussion

In the present article, a previously developed model for hybrid rolling bearing life calculations has been adapted to reflect the performance of a tougher material used in high-speed and high-load combinations, proper of some important applications like spindle bearings. For this, two full endurance tests were necessary, one for the baseline steel ASTM 52100 and the other for the special steel PM1. Besides this a minimum-life endurance test was necessary at the application conditions to set the limits of the model and bearing applicability as a combination of load and speed in order to avoid seizure.

The results show that it is possible to adapt the hybrid bearing life model to account for the particular performance of different steels with the limitations load-speed also given by the particular materials considered in the bearing design. As discussed in [9, 11] the PM1 steel has high thermal conductivity and high fatigue strength, which make it a very successful material for conditions of high speed and high load as the ones pursued in these applications if compared with the earlier results for hybrid bearings [8] as shown in Fig. 7. In the performed fatigue life tests the superior performance of PM1 is also evident.

7 Conclusion

From the investigations performed in this work, the following conclusions can be drawn:

1. The modified hybrid bearing life model now reflects the superior fatigue performance of the PM1 steel as shown by the two endurance tests described above.
2. The steel PM1 also shows superior seizure failure resistance in conditions of 3 GPa pressure and 2.5 million nd_m speed proper of spindle applications, if compared with the hybrid bearing limits earlier obtained in [8].

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Nomenclature

a	Contact semi-axis in transverse direction	[mm]
A	Surface integration area	[m ²]
$\bar{A}$	Damage integral constant for the volume	[-]
$\bar{B}$	Damage integral constant for the surface	[-]
c	Exponent in the bearing life equation, and constant	[-]
C	Dynamic load rating (subsurface)	[N]
d_m	Mean diameter of the bearing	[mm]
e	Exponent in the bearing life equation (standardised Weibull slope)	[-]
h	Exponent in the bearing life equation	[-]
L_{10}	Bearing life rating with 90% reliability	[Mrev]

$L_{10,5}$	Lower Bound of the 90% two-sided confidence interval on L_{10}	[Mrev]
$L_{10,95}$	Upper Bound of the 90% two-sided confidence interval on L_{10}	[Mrev]
$L_{10,50}$	Median L_{10} Estimate	[Mrev]
L_{50}	Bearing life rating with 50% reliability	[Mrev]
n	Bearing rotational speed	[rpm]
P	Equivalent load in the bearing	[N]
p	Exponent in the bearing life equation	[-]
u	Number of stress cycles per revolution	[-]
S	Reliability	[-]
V_v	Integration volume	[m ³]
z	Coordinate (depth from surface)	[m]

Greek symbols

α	Raceway contact angle	[deg]
η_{lub}	Lubrication quality factor for life,	[-]
η_{cont}	Contamination factor for life	[-]
η_{env}	Environmental factor for life, $\eta_{env} = \eta_{lub} \eta_{cont}$	[-]
κ	Viscosity ratio in the bearing	[-]
σ	Fatigue generating stress	[Pa]
τ	Fatigue stress in AFC life (orthogonal x-z shear stress), τ_{xz}	[Pa]
τ_0	Maximum alternating shear stress amplitude	[Pa]
τ_u	Fatigue limit stress	[Pa]

Subscripts

v	Refers to volume
s	Refers to surface

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